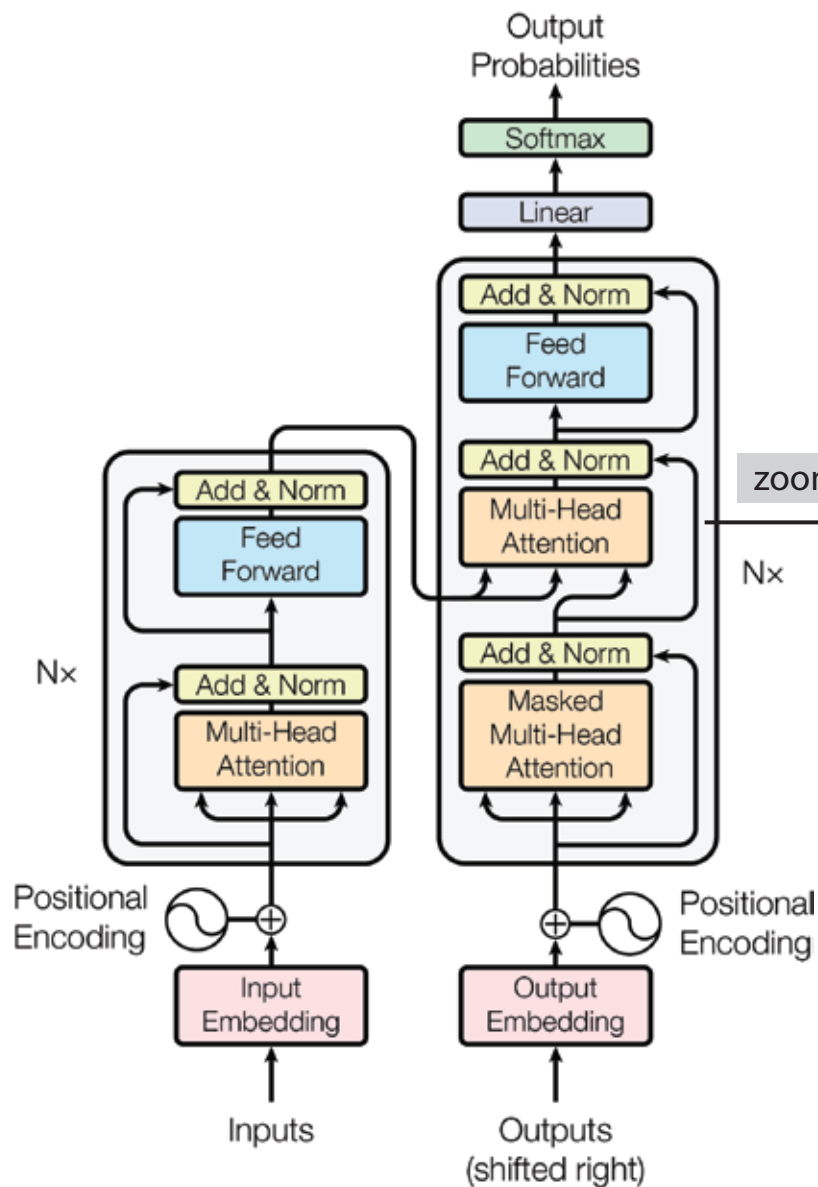
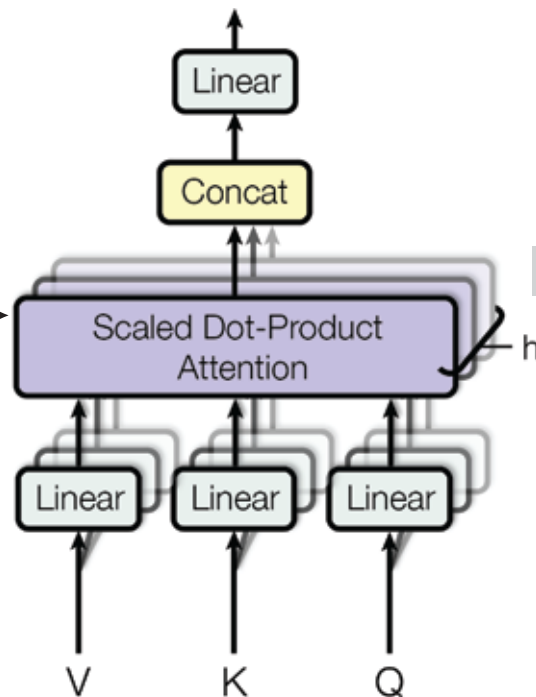
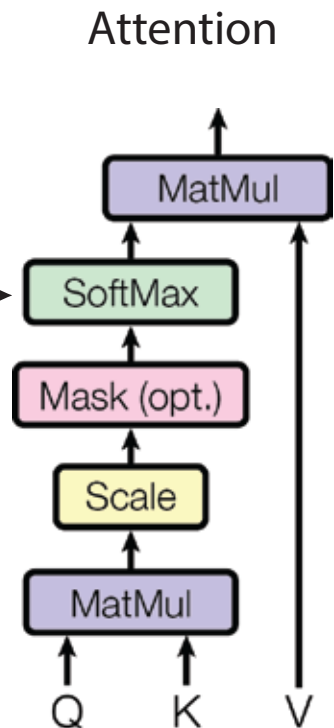




Multi-Head Attention



Scaled Dot-Product Attention



$$\text{Attention}(Q, K, V) = \text{softmax} \left(\frac{QK^T}{\sqrt{d_k}} \right) V$$

$$\text{MultiHead}(Q, K, V) = \text{Concat}(\text{head}_1, \dots, \text{head}_h) W^O$$

$$\text{where } \text{head}_i = \text{Attention} \left(\underbrace{Q W_i^Q}_{XW_i^Q}, \underbrace{K W_i^K}_{XW_i^K}, \underbrace{V W_i^V}_{XW_i^V} \right)$$

Encoder

Decoder